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Search Committee
Department of Economics and Business
Colorado School of Mines
Golden, CO

Dear Members of the Search Committee,

I am writing to apply for the tenure track Assistant Professor position in Business Analytics and Operations Research at the Colorado School of Mines. I am a Ph.D. candidate in the Department of Economics at Johns Hopkins University (advisors: Christopher D. Carroll, Nicholas W. Papageorge, and Stelios Fourakis), and I expect to receive my degree in May 2026. My primary specializations are macroeconomic modeling, wealth and income dynamics, and computational economics.

My research agenda focuses on the ability of macroeconomic models to generate empirically realistic distributions of wealth and income over the life cycle. A key driver of that work is modeling heterogeneity in the rate of return on assets, and understanding how return dispersion contributes to portfolio accumulation across households.

In my job market paper, I calibrate a life cycle heterogeneous agent model to match wealth moments from the Survey of Consumer Finances by allowing households to differ in their portfolio returns. The model helps explain observed wealth concentration and return dispersion. In a related research project using the Health and Retirement Study, I examine how trust influences both income growth and asset returns. I find a hump shaped relationship for trust and returns (consistent with the literature) and characterize the distribution of the persistent component of returns to net worth. These experiences have equipped me to frame research strategies and design modeling approaches, execute econometric and simulation analysis with large micro data sets, and translate results into clear and rigorous written deliverables for both academic and practitioner audiences.

At a technical level, solving heterogeneous agent models requires rigorous understanding of dynamic stochastic optimization and decision making under uncertainty. These models are high dimensional optimization problems with forward looking agents responding to stochastic income, return, and policy processes. Their solution involves value function iteration, simulation, and numerical approximation techniques that are closely related to optimization under uncertainty and simulation methods in operations research. This methodological core aligns naturally with Mines' emphasis on analytics, optimization, and decision support in complex systems.

I am particularly drawn to Mines because of its strong interdisciplinary focus on energy systems, natural resources, and sustainability. Questions about investment under uncertainty, risk allocation, and long horizon planning are central both to my research and to Mines' mission. I would welcome the opportunity to contribute to quantitative business programs while collaborating with faculty working on energy markets, resource economics, and integrated systems modeling. I am enthusiastic about relocating to Golden and engaging fully in the in person academic environment.

A tenure track appointment at Mines would provide an excellent platform to expand my research agenda in computational economics and applied optimization, while pursuing external funding and interdisciplinary collaboration.

I believe I would be a strong fit for this position because:

1. My research is grounded in dynamic stochastic optimization, simulation, and quantitative modeling under uncertainty, which directly supports business analytics and operations research instruction.
2. I have experience teaching and communicating complex quantitative concepts in accessible ways at the undergraduate and graduate levels.
3. My research agenda connects naturally to Mines' priorities in energy, natural resources, and sustainable economic development.

Thank you very much for your consideration. I would welcome the opportunity to discuss how my research and teaching interests align with the goals of the Department of Economics and Business at Mines.

Sincerely,

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